

Deep Learning Lab 3 Report

Air Quality Index Classification using CNN and Transfer Learning

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1 Objective

The objective of this lab is to classify Air Quality Index (AQI) levels from environmental images using deep learning models.

The lab explores two different approaches:

- A custom Convolutional Neural Network (CNN)
- Transfer learning using a pre-trained ResNet50 model

The performance of both models is compared using classification metrics.

2 Dataset

The dataset contains images labeled according to their Air Quality Index category.

Total dataset size:

6000 images

2.1 Dataset Split

Dataset Split	Images	Percentage
Training	4200	70%
Validation	900	15%
Test	900	15%

2.2 AQI Categories

The dataset contains six AQI classes:

- a_Good
- b_Moderate
- c_Unhealthy_for_Sensitive_Groups

- d_Unhealthy
- e_Very_Unhealthy
- f_Severe

3 Baseline Model: Custom CNN

A custom CNN model was implemented for image classification.

The model consists of convolutional layers followed by fully connected layers.

3.1 Training Results

Epoch	Train Accuracy	Validation Accuracy
1	0.355	0.431
2	0.475	0.541
3	0.469	0.506
4	0.488	0.497
5	0.393	0.410

3.2 Test Performance

Overall accuracy:

$$Accuracy = 0.43$$

Class	Precision	Recall	F1 Score
a_Good	0.39	0.57	0.46
b_Moderate	0.43	0.37	0.40
c_Unhealthy_for_Sensitive_Groups	0.53	0.14	0.22
d_Unhealthy	0.79	0.17	0.28
e_Very_Unhealthy	0.34	0.75	0.47
f_Severe	0.56	0.57	0.57

4 Transfer Learning using ResNet50

To improve performance, a pre-trained ResNet50 model was used with transfer learning.

The final fully connected layer was modified to output six AQI classes.

4.1 Training Results

Epoch	Train Accuracy	Validation Accuracy
1	0.682	0.697
2	0.718	0.736
3	0.743	0.738
4	0.766	0.758
5	0.772	0.783

4.2 Test Performance

Overall accuracy:

$$Accuracy = 0.75$$

Class	Precision	Recall	F1 Score
a_Good	0.77	0.74	0.76
b_Moderate	0.66	0.71	0.69
c_Unhealthy_for_Sensitive_Groups	0.84	0.37	0.52
d_Unhealthy	0.63	0.85	0.72
e_Very_Unhealthy	0.79	0.87	0.83
f_Severe	0.89	0.95	0.92

5 Model Comparison

Model	Test Accuracy
Custom CNN	0.43
ResNet50 (Transfer Learning)	0.75

6 Conclusion

This lab explored image classification using convolutional neural networks.

Key observations:

- The custom CNN achieved moderate performance with an accuracy of 43%.
- Transfer learning significantly improved performance.
- The ResNet50 model achieved 75% accuracy on the test set.
- Pre-trained models are highly effective when working with relatively small datasets.

The experiment demonstrates the effectiveness of transfer learning in image classification tasks.